

Are galaxies in compact groups special?

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ABSTRACT

We investigate the properties of galaxies in compact groups (CGs) and compare them to a control sample of galaxies.

Key words. galaxies: clusters: general – catalogs

1. Introduction

2. Data

2.1. Samples

We use the compact groups and control samples we built in our previous article [Tricottet et al. \(2025\)](#). We nevertheless briefly recall here how they were built. [Explain.](#) Contrarily to our previous article, though, we remove CGs that we classified as *split* following [Zheng & Shen \(2021\)](#). In this process, our initial compact group sample of 781 groups reduced to 62. We also remove groups built from Lim group 3688 from our Control_{4B} and Control_{4C} samples. This group has six galaxies having a redshift between 0.032 and 0.034, and one having a redshift of 0.048. This group seems spurious, and the outlier galaxy, although not being the BGG, is selected both in our original Control_{4B} and Control_{4C}, influencing the velocity dispersion of the group significantly (1979 and 1981 km s⁻¹, respectively).

We use the SDSS DR 16 to retrieve masses, star formation rates (SFRs) and morphologies of galaxies assessed in Galaxy Zoo ([Lintott et al. \(2011\)](#)). We specifically extracted fields `sfr_tot_p50`, `specsfr_tot_p50` and `lgm_tot_p50` from the `galSpecExtra` table, and `p_el_debiased` and `p_cs_debiased` from the `zooSpec` table.

2.2. *sSFR*

To build a general star formation classification that could be uniformly applied to all our samples, we extracted galaxies from the SDSS DR 16 through the query displayed in listing 1.

```
p.petroMag_r - p.extinction_r as
r_obs,
p.petroMag_u - p.extinction_u as
u_obs,
p.petroMag_g - p.extinction_g as
g_obs,
p.petroMag_i - p.extinction_i as
i_obs,
p.petroMag_z - p.extinction_z as
z_obs,
p.objID,
g.sfr_tot_p50, g.specsfr_tot_p50, g.
lgm_tot_p50,
1.h_alpha_eqw, 1.h_beta_eqw, 1.
oiii_5007_eqw, 1.nii_6584_eqw,
1.h_alpha_flux, 1.h_beta_flux, 1.
oiii_5007_flux, 1.nii_6584_flux,
z.p_el_debiased AS p_E,
z.p_cs_debiased AS p_S
FROM SpecObj AS s
JOIN PhotoObj AS p ON s.bestObjID = p
.objID
JOIN galSpecExtra as g ON s.specObjID
= g.specObjID
JOIN galSpecLine as l ON s.specObjID
= l.specObjID
JOIN zooSpec AS z ON s.specObjID = z.
specObjID
WHERE s.z BETWEEN 0.005 AND 0.0452
AND (p.petroMag_r - p.extinction_r <=
17.77)
AND s.class = 'GALAXY'
AND g.lgm_tot_p50 > -1000
```

Listing 1: Query used for selecting spectral & photometric data from the SDSS

```
SELECT
s.specObjID,
s.z,
```

We select non-AGN galaxies by first placing them on the classical BPT diagnostic of [Veilleux & Osterbrock \(1987\)](#) and requiring measured values of both emission-line ratios

$\log_{10}([\text{N II}]\lambda 6584/\text{H}\alpha)$ and $\log_{10}([\text{O III}]\lambda 5007/\text{H}\beta)$.

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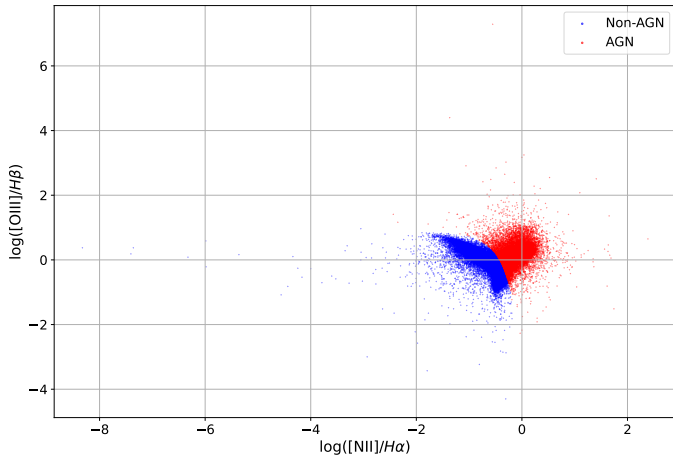


Fig. 1: BPT diagram showing our separation between ordinary and AGN galaxies

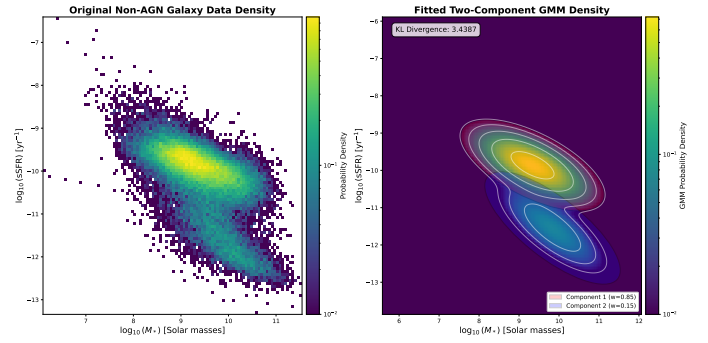


Fig. 2

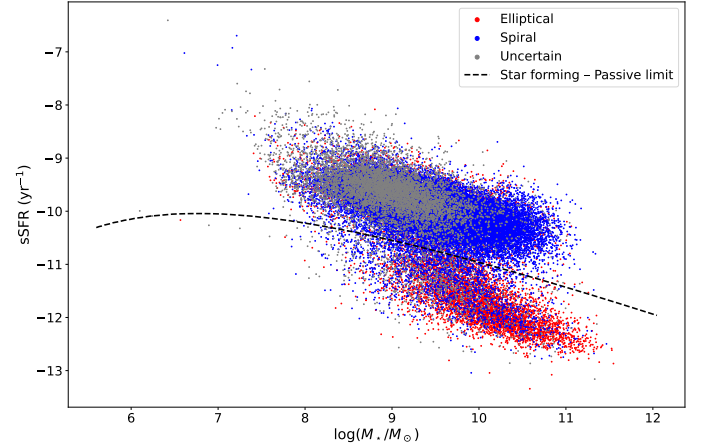


Fig. 3: sSFR vs mass, star-forming/passive limit and Zoo morphologies for the SDSS control sample, excluding quenched galaxies.

A galaxy is flagged as an AGN if it satisfies either

$$\log_{10} \frac{[\text{NII}]}{\text{H}\alpha} > 0,$$

or if it lies above the empirical demarcation of Kauffmann et al. (2003):

$$\log_{10} \frac{[\text{OIII}]}{\text{H}\beta} > \frac{0.61}{\log_{10}([\text{NII}]/\text{H}\alpha) - 0.05} + 1.3,$$

in which case it is removed from the non-AGN sample. All remaining galaxies—including those with missing line ratios—are retained as non-AGN. The non-AGN selection process is shown on figure 1.

We identify the galaxies having a `specs_f_tot_p50` of -9999 in the SDSS as quenched (for graphic representation purposes, we give it here an arbitrary low value of $10^{-14.0} \text{ Gyr}^{-1}$). We then consider the other ones and sort them between the star-forming and passive populations using a two-component Gaussian Mixture Model (GMM) in the $\log M_*$ – $\log \text{sSFR}$ plane. Its parameters determined by minimizing the Kullback–Leibler divergence between the empirical and model densities (Kullback & Leibler (1951); McLachlan & Peel (2000)). For any trial parameter vector θ , we reconstruct the mixture means $\{\mu_i\}$, covariances $\{\Sigma_i\}$, and weights $\{w_i\}$ via a Cholesky-like decomposition and logistic mapping, optionally constraining the second component’s mean to isolate the passive population (Dempster et al. (1977)). The KL divergence is estimated by constructing a 2D histogram of the data and summing $p_{\text{data}} \ln(p_{\text{data}}/p_{\text{GMM}})$, with a small regularization ϵ to avoid singularities (Tojeiro et al. (2009); Bisigello et al. (2018)). We optimize θ via L-BFGS-B (with fallback to Nelder–Mead) over multiple guided and random initializations, selecting the solution with lowest divergence (Pedregosa et al. (2011); Bishop (2006)). Initial means are seeded by a median split in $\log \text{sSFR}$ to improve convergence and robustness against local minima (Rousseeuw (1987)). In the final fit, one component naturally aligns with the high-sSFR “star-forming sequence,” while the other captures the low-sSFR “passive” cloud, providing a data-driven division that agrees with previous multi-Gaussian decompositions of the $\text{SFR} - M_*$ plane (Wuyts et al. (2011); Hahn et al. (2019)). The decision boundaries, defined as the

loci where the posterior probabilities of adjacent components are equal, provide an objective criterion for delineating star-forming from passive galaxies.

Add relevant GM parameters found by the algorithm.

The sSFR status for each sample is shown in table 1. There is no significant differences of for Star forming proportion between CG_4 and the control samples (probabilities estimated using Fisher exact test are 6.5×10^{-1} versus Control_{4B} sample, 2.5×10^{-1} versus Control_{4C} and 1.8×10^{-2} versus RG₄).

Restrictions of the comparison to BGGs and satellites only are also shown in table 1. The p-values found from two-sided Fisher exact test are 0.58 against Control_{4B}, 0.59 against Control_{4C} and 0.37 against RG₄. There is thus no significant difference in the fraction of star-forming BGGs nor satellites between CG_4 and our control samples either.

2.3. Morphology

Morphologies are determined via the Galaxy Zoo citizen-science decision tree, in which each galaxy image receives multiple independent volunteer classifications that are aggregated into debiased vote fractions Lintott et al. (2011). Those fractions are provided in the `zooSpecz` table of the SDSS databasethrough the `p_el_debiased` and `p_cs_debiased` columns. We attribute to galaxies the morphology “elliptical” if the `p_el_debiased` value is greater than 0.5, and “spiral” if the `p_cs_debiased` value is greater

Sample		Quenched	Passive	Starforming
CG ₄		18 (7.3 %)	146 (58.9 %)	84 (33.9 %)
	<i>BGG</i>	6 (9.7 %)	45 (72.6 %)	11 (17.7 %)
	<i>Satellites</i>	12 (6.5 %)	101 (54.3 %)	73 (39.2 %)
Control _{4B}		181 (6.5 %)	1596 (57.1 %)	1019 (36.4 %)
	<i>BGG</i>	58 (8.3 %)	534 (76.4 %)	107 (15.3 %)
	<i>Satellites</i>	123 (5.9 %)	1062 (50.6 %)	912 (43.5 %)
Control _{4C}		191 (6.3 %)	1619 (53.8 %)	1198 (39.8 %)
	<i>BGG</i>	65 (8.6 %)	569 (75.7 %)	118 (15.7 %)
	<i>Satellites</i>	126 (5.6 %)	1050 (46.5 %)	1080 (47.9 %)
RG ₄		5 (2.2 %)	101 (45.1 %)	118 (52.7 %)
	<i>BGG</i>	3 (5.4 %)	38 (67.9 %)	15 (26.8 %)
	<i>Satellites</i>	2 (1.2 %)	63 (37.5 %)	103 (61.3 %)
<i>SDSS selection</i>		634 (1.2 %)	7705 (14.7 %)	44192 (84.1 %)

Table 1: Number of galaxies in each sSFR status for each sample.

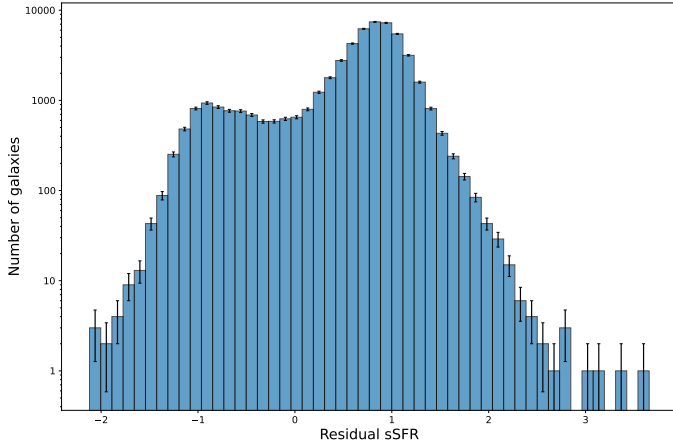
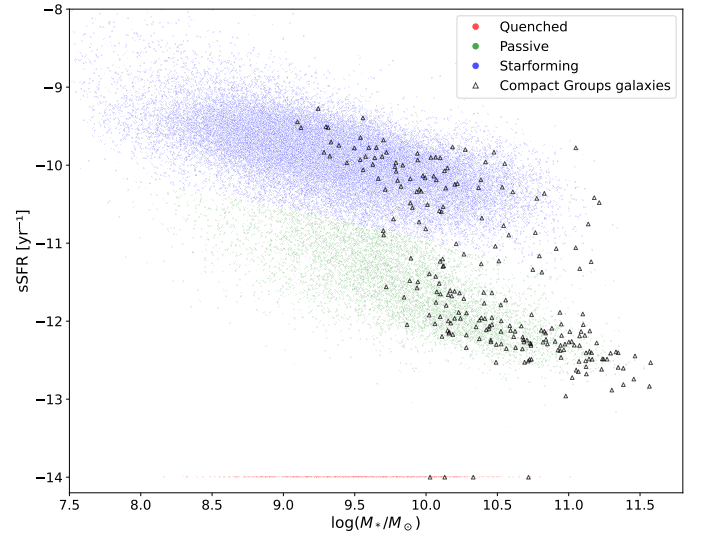


Fig. 4


 Fig. 5: sSFR and morphologies of galaxies in CG₄ and our SDSS selection. Quenched galaxies are represented at an arbitrary low value of $10^{-14.0} \text{ Gyr}^{-1}$.

than 0.5. Other cases are considered as “uncertain”. The morphologies found for each sample are shown in table 2. Barnard two-sided exact tests were performed to compare the fraction of each morphology between CG₄ and each control sample. The p-values found are for ellipticals and for spirals versus Control_{4B} sample, and versus Control_{4C} and and versus RG₄. Morphologies of CG₄, which we identified as cores of richer groups in Tricottet et al. (2025), have morphologies significantly different from ordinary groups of four members. Figure 5 displays the sSFR against masses for galaxies from the SDSS selection and for CG₄s galaxies.

2.4. Morphology vs sSFR

Starforming CG₄ are composed of of their galaxies classified as spiral (% once galaxies having uncertain morphology removed) and (% once galaxies having uncertain morphology removed) elliptical. These figures are (%) spiral and (%) elliptical galaxies for Control_{4B}, (%) spiral and (%) elliptical galaxies for Control_{4C} and (%) spiral and (%) elliptical galaxies for RG₄. The probability that those fractions come from the same distribution is estimated from the as against Control_{4B}, against Control_{4C} and against RG₄. There is thus neither significant excess nor lack of starforming elliptical galaxies in CG₄ compared to all our control samples.

3. sSFR

3.1. Main sequence

We model the star-forming main sequence as a power law in stellar mass, to various orders (1 to 4) - cf figure 9. We then choose to use order 2 to compute the residuals of each star-forming galaxy to the main sequence, and plot their histograms in figure 10.

The median main-sequence residual of CG₄ galaxies is significantly different

Sample	Elliptical	Spiral	Uncertain
CG ₄	124 (%)	86 (%)	38 (%)
Control _{4B}	1107 (%)	1268 (%)	421 (%)
Control _{4C}	1215 (%)	1251 (%)	542 (%)
RG ₄	60 (%)	129 (%)	35 (%)
<i>SDSS selection</i>	<i>10091 (%)</i>	<i>34715 (%)</i>	<i>7725 (%)</i>

Table 2: Number of galaxies of each morphology for each sample.

Sample	$\langle \log sSFR \rangle$		$\langle M_r \rangle$		$\langle \log(M_\star/M_\odot) \rangle$	
	Passive	Star-forming	Passive	Star-forming	Passive	Star-forming
CG4	-12.08	-10.19	-21.16	-20.4	10.64	10.05
Control4B	-12.1	-10.31	-21.52	-20.75	10.79	10.24
Control4C	-11.92	-10.16	-20.81	-19.93	10.47	9.82
RG4	-11.91	-10.16	-21.11	-20.32	10.62	9.97

Table 3: Median values of galaxy properties for passive and star-forming populations in each sample. Error bars are computed via bootstrapping (10 000 iterations).

Sample	sSFR (yr ⁻¹)		M_r		$\log(M_\star/M_\odot)$	
	Passive BGG	Star-forming BGG	Passive BGG	Star-forming BGG	Passive BGG	Star-forming BGG
CG4	-11.53 ± 0.09	-10.85 ± 0.29	-20.93 ± 0.11	-20.77 ± 0.32	10.46 ± 0.06	10.32 ± 0.12
Control4B	-11.53 ± 0.03	-10.86 ± 0.06	-21.25 ± 0.03	-21.12 ± 0.07	10.61 ± 0.02	10.46 ± 0.04
Control4C	-11.3 ± 0.03	-10.62 ± 0.06	-20.42 ± 0.05	-20.55 ± 0.08	10.22 ± 0.02	10.15 ± 0.06
RG4	-11.14 ± 0.19	-10.58 ± 0.17	-20.69 ± 0.14	-20.74 ± 0.28	10.32 ± 0.07	10.19 ± 0.12

Table 4: Median values of various quantities for galaxies according to the fertility of their BGG (BGG included). Error bars are computed via bootstrapping (10 000 iterations).

from that of Control4B, with a median offset of $\Delta = 0.056$ ($0.029 \leq \Delta \leq 0.097$, bootstrap), corresponding to a p -value of $p = 0.059$.

The median main-sequence residual of CG4 galaxies is significantly different

from that of Control4C, with a median offset of $\Delta = 0.082$ ($0.063 \leq \Delta \leq 0.122$, bootstrap), corresponding to a p -value of $p = 0.008$.

The median main-sequence residual of CG4 galaxies is not significantly different

from that of RG4, with a median offset of $\Delta = 0.046$ ($-0.037 \leq \Delta \leq 0.097$, bootstrap), corresponding to a p -value of $p = 0.548$.

A direct rank-sum comparison of the star-forming main-sequence offsets only identifies significant differences between CG₄ and

Control_{4C}, whose median offset is -0.060 with respect to the fitted sequence ($p = 0.03$).

Within CG₄, the star-forming main-sequence offsets of the Embedded, Isolated and Predom classes remain statistically consistent with one another.

3.2. Distance to the BGG

We also compare the sSFR of satellite galaxies with their distance to the BGG, expressed both in projected kpc and in units of $\langle R_{ij} \rangle$.

Significant monotonic trends are detected in the following cases.

In CG₄, sSFR correlates with projected distance to the BGG with Spearman coefficient $\rho_s = 0.24$ for 177 satellites

($p = 1.2 \times 10^{-3}$) and Pearson coefficient $r = 0.21$ ($p = 4.1 \times 10^{-3}$).

In Control_{4B}, sSFR correlates with distance to the BGG normalized by $\langle R_{ij} \rangle$ with Spearman coefficient $\rho_s = 0.08$ for 1991 satellites ($p = 3.6 \times 10^{-4}$) and Pearson coefficient $r = 0.08$ ($p = 2.3 \times 10^{-4}$).

In Control_{4C}, sSFR correlates with projected distance to the BGG with Spearman coefficient $\rho_s = 0.25$ for 2143 satellites ($p = < 10^{-6}$) and Pearson coefficient $r = 0.20$ ($p = < 10^{-6}$).

In Control_{4C}, sSFR correlates with distance to the BGG normalized by $\langle R_{ij} \rangle$ with Spearman coefficient $\rho_s = 0.10$ for 2143 satellites ($p = 4.6 \times 10^{-6}$) and Pearson coefficient $r = 0.09$ ($p = 1.8 \times 10^{-5}$).

3.3. Dominance

We split groups into two categories: dominated groups, for which the BGG luminosity fraction exceeds our threshold of 0.6, and non-dominated groups. Among the group properties explored in this work, the quantities showing at least one significant dominated/non-dominated difference are

$$\begin{aligned} &\langle R_{ij} \rangle \text{ (kpc)}, \\ &\Delta_{\text{BGG-cen}} / \langle R_{ij} \rangle, \\ &\sigma_v \text{ (km s}^{-1}\text{)}, \\ &\Delta V_{\text{BGG}} / \sigma_v \text{ and} \\ &L_{\text{group}} (L_\odot). \end{aligned}$$

In CG₄, no dominated/non-dominated difference remains significant after the FDR correction.

Sample	sSFR (yr^{-1})		M_r	Star-forming BGG	$\log(M_\star/M_\odot)$	
	Passive BGG	Star-forming BGG	Passive BGG	Star-forming BGG	Passive BGG	Star-forming BGG
CG4	-11.23 ± 0.16	-10.92 ± 0.48	-20.45 ± 0.13	-20.34 ± 0.18	10.22 ± 0.05	10.09 ± 0.07
Control4B	-11.26 ± 0.06	-10.91 ± 0.13	-20.9 ± 0.03	-20.75 ± 0.06	10.43 ± 0.02	10.3 ± 0.03
Control4C	-10.96 ± 0.05	-10.6 ± 0.07	-19.8 ± 0.04	-20.02 ± 0.10	9.91 ± 0.02	9.89 ± 0.07
RG4	-10.82 ± 0.19	-10.61 ± 0.22	-20.23 ± 0.16	-20.3 ± 0.13	10.09 ± 0.08	10 ± 0.13

Table 5: Median values of various quantities for galaxies according to the fertility of their BGG (satellites only). Error bars are computed via bootstrapping (10 000 iterations).

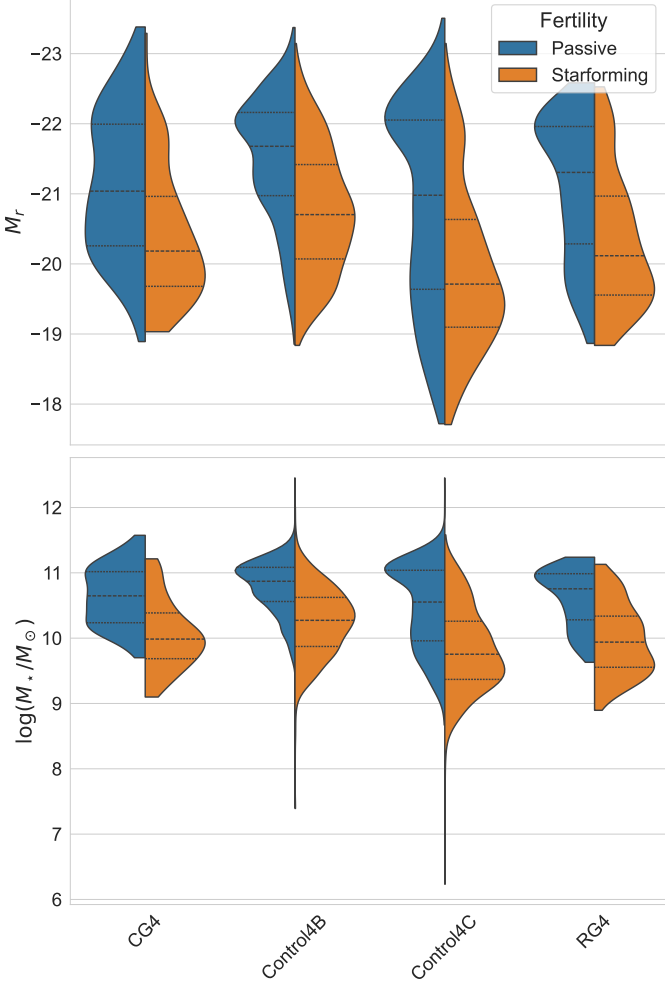


Fig. 6: Distribution various quantities for galaxies according their fertility.

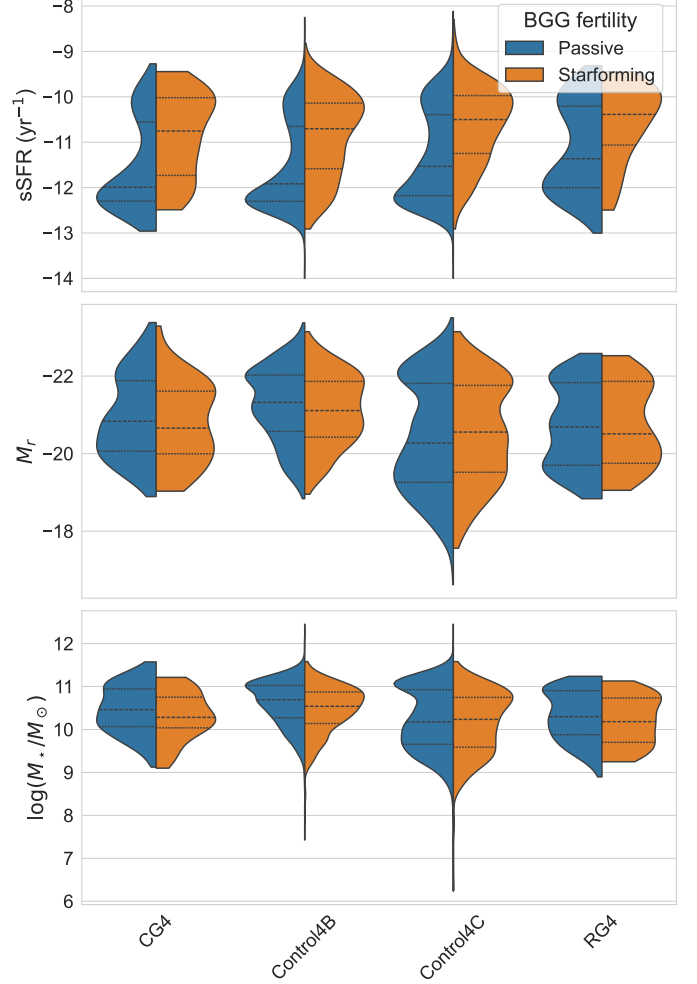


Fig. 7: Distribution various quantities for galaxies according to the fertility of their BGG (BGG included).

In Control_{4B}, the strongest corrected dominated/non-dominated separation concerns $L_{\text{group}} (L_\odot)$: the dominated groups have a median value of $9.1\text{e}+10$, compared to $1.28\text{e}+11$ for the non-dominated ones ($p_{\text{adj}} = < 10^{-6}$; Cliff's $\delta = -0.55$).

In Control_{4C}, the strongest corrected dominated/non-dominated separation concerns $L_{\text{group}} (L_\odot)$: the dominated groups have a median value of $8.49\text{e}+10$, compared to $9.54\text{e}+10$ for the non-dominated ones ($p_{\text{adj}} = < 10^{-6}$; Cliff's $\delta = -0.25$).

In RG₄, the strongest corrected dominated/non-dominated separation concerns $L_{\text{group}} (L_\odot)$: the dominated groups have a median value of $7.09\text{e}+10$, compared to

$9.82\text{e}+10$ for the non-dominated ones ($p_{\text{adj}} = 7.8 \times 10^{-5}$; Cliff's $\delta = -0.59$).

The group spiral fraction also correlates significantly with the group-scale observables in a subset of the samples.

For Control_{4C} dominated groups, S_{frac} correlates with $\langle R_{ij} \rangle$ (kpc) ($\rho_S = 0.38$, 527 groups, $p = < 10^{-6}$).

For Control_{4C} non-dominated groups, S_{frac} correlates with $\langle R_{ij} \rangle$ (kpc) ($\rho_S = 0.35$, 224 groups, $p = < 10^{-6}$).

For Control_{4C} dominated groups, S_{frac} correlates with σ_v (km s^{-1}) ($\rho_S = -0.23$, 527 groups, $p = < 10^{-6}$).

For Control_{4C} dominated groups, S_{frac} correlates with $L_{\text{group}} (L_\odot)$ ($\rho_S = -0.15$, 527 groups, $p = 8.4 \times 10^{-4}$).

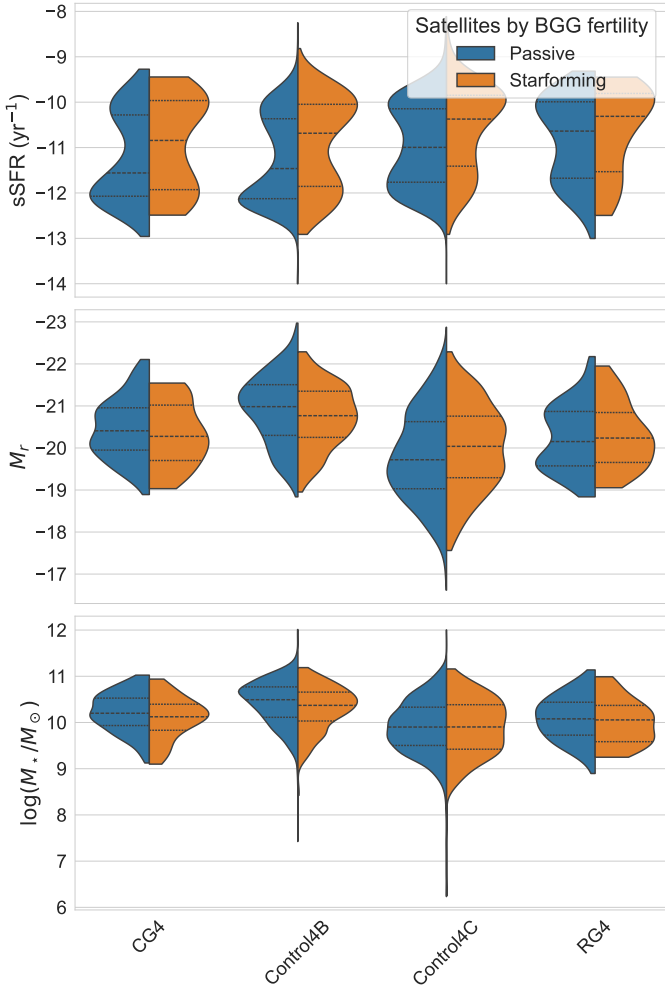


Fig. 8: Distribution various quantities for galaxies according to the fertility of their BGG (BGG excluded).

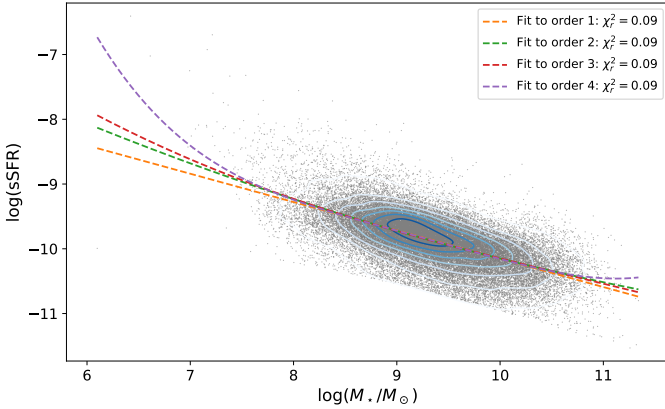


Fig. 9

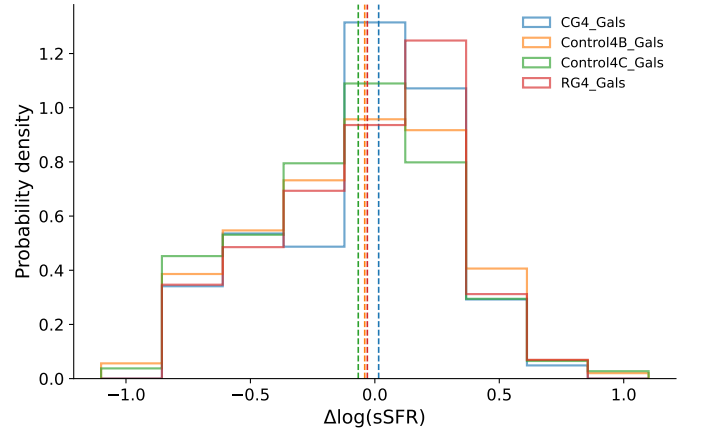


Fig. 10

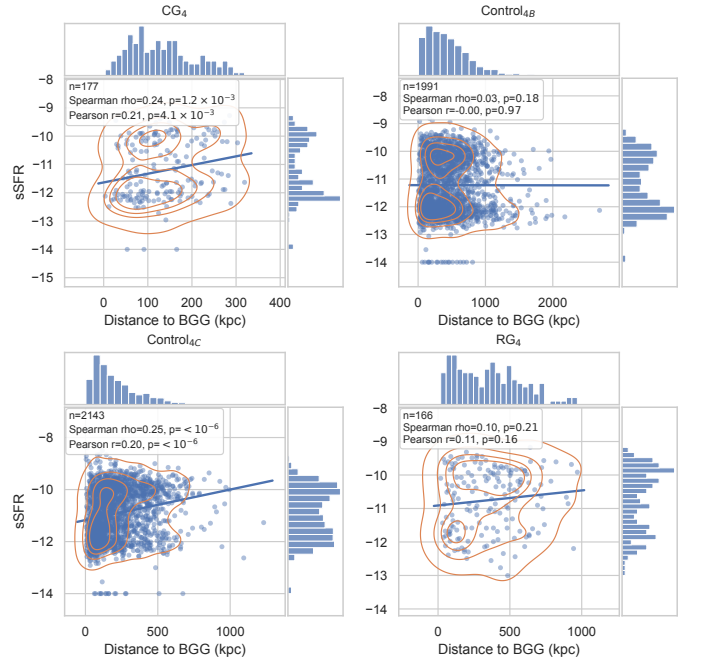


Fig. 11: sSFR as a function of projected distance to the BGG, in kpc, for satellite galaxies in each sample.

For CG₄ non-dominated groups, S_{frac} correlates with $\Delta V_{\text{BGG}}/\sigma_v$ ($\rho_s = -0.53$, 26 groups, $p = 5.6 \times 10^{-3}$).

For Control_{4C} non-dominated groups, S_{frac} correlates with L_{group} ($L_{\odot}$) ($\rho_s = -0.15$, 224 groups, $p = 0.02$).

For Control_{4B} non-dominated groups, S_{frac} correlates with σ_v (km s^{-1}) ($\rho_s = -0.09$, 504 groups, $p = 0.03$).

For Control_{4B} non-dominated groups, S_{frac} correlates with $\langle R_{ij} \rangle$ (kpc) ($\rho_s = 0.09$, 504 groups, $p = 0.04$).

For CG₄ non-dominated groups, S_{frac} correlates with $\langle R_{ij} \rangle$ (kpc) ($\rho_s = 0.40$, 26 groups, $p = 0.04$).

3.4. Morphology and domination

When we further split the samples by domination and by the morphology of the BGG, significant effects only appear in

260 For Control_{4B} non-dominated groups, S_{frac} correlates with L_{group} ($L_{\odot}$) ($\rho_s = -0.14$, 504 groups, $p = 1.4 \times 10^{-3}$).

For CG₄ dominated groups, S_{frac} correlates with L_{group} ($L_{\odot}$) ($\rho_s = -0.50$, 36 groups, $p = 1.9 \times 10^{-3}$).

For Control_{4C} non-dominated groups, S_{frac} correlates with σ_v (km s^{-1}) ($\rho_s = -0.20$, 224 groups, $p = 2.6 \times 10^{-3}$).

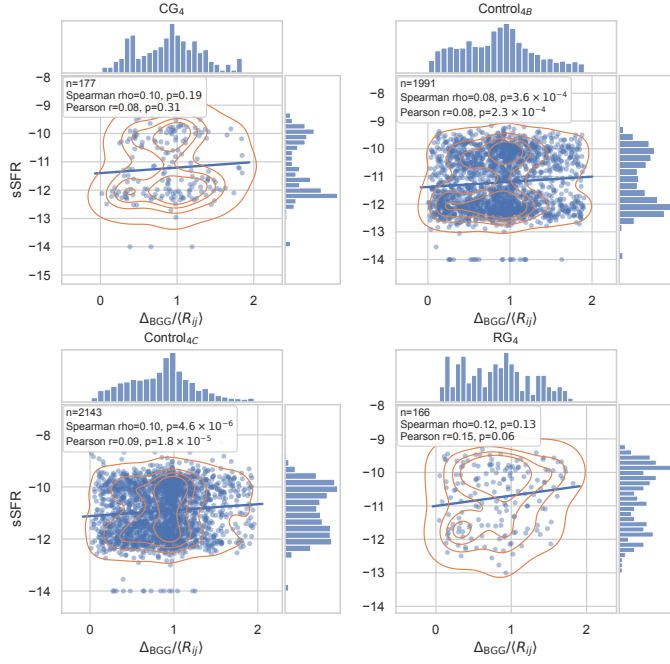


Fig. 12: sSFR as a function of projected distance to the BGG normalized by $\langle R_{ij} \rangle$ for satellite galaxies in each sample.

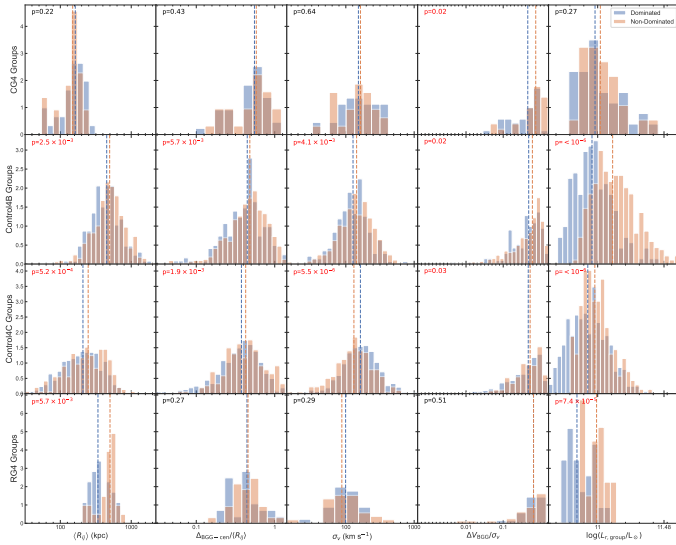


Fig. 13: Distributions of the main group-scale observables in dominated and non-dominated systems.

Control_{4B} dominated groups, where the median satellite spiral fraction is 1.00 for spiral BGGs and 0.67 for elliptical BGGs. The corresponding group-level odds ratio is 1.84 ($p = 0.01$), while the satellite-level clustered fit gives an odds ratio of 1.84 ($p = 0.02$);

Control_{4C} dominated groups, where the median satellite spiral fraction is 0.67 for spiral BGGs and 0.50 for elliptical BGGs. The corresponding group-level odds ratio is 1.84 ($p = 2.1 \times 10^{-5}$), while the satellite-level clustered fit gives an odds ratio of 1.84 ($p = 2.7 \times 10^{-5}$).

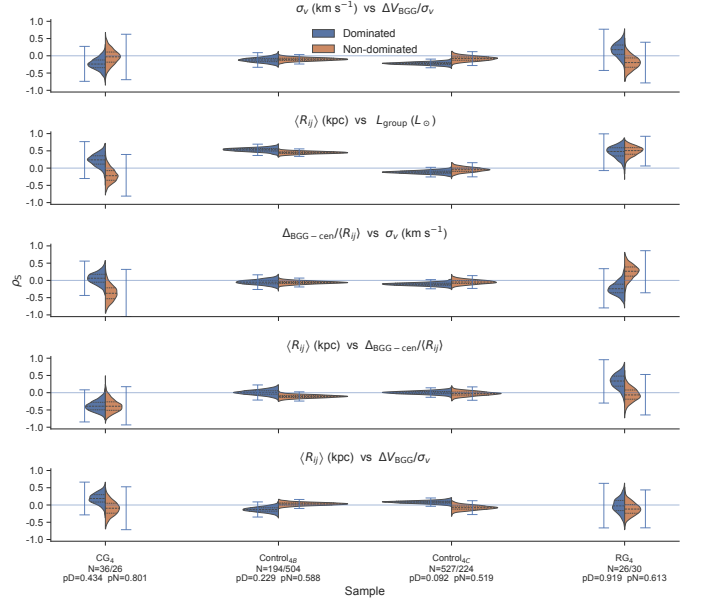


Fig. 14: Bootstrap distributions of the Spearman coefficients for the group-property pairs selected by the XOR domination criterion.

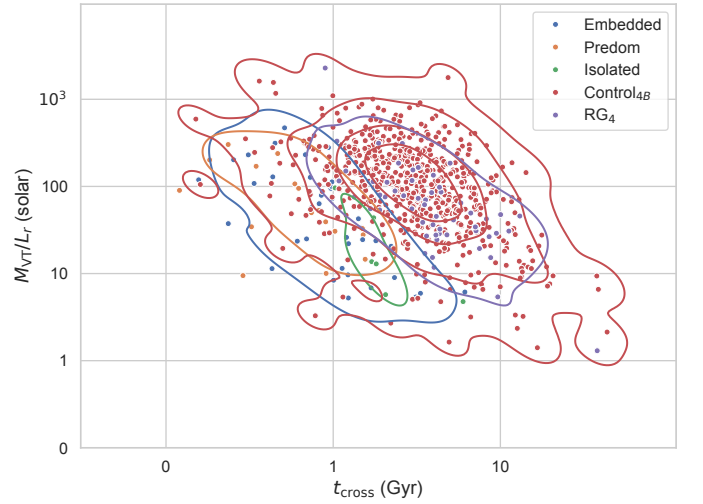


Fig. 15: Crossing time versus virial mass-to-light ratio for the compact groups and their controls.

4. Group Dynamics

The crossing-time exploration quantifies the trends illustrated in figures 15 to 17.

For $\log(M_{VT}/L_r)$, the global Spearman coefficient with $\log t_{\text{cross}}$ is $\rho_S = -0.39$ based on 816 groups ($p = < 10^{-6}$).

For $\log(M_{VT})$, the global Spearman coefficient with $\log t_{\text{cross}}$ is $\rho_S = -0.33$ based on 816 groups ($p = < 10^{-6}$).

For $\log(L_{\text{group}})$, the global Spearman coefficient with $\log t_{\text{cross}}$ is $\rho_S = 0.08$ based on 816 groups ($p = 0.03$).

5. Colour

All galaxies are redder when they have a higher stellar mass, as per Alpaslan et al. (2015) - cf figure ?? . Stellar mass is therefore a confounding factor when comparing the colors

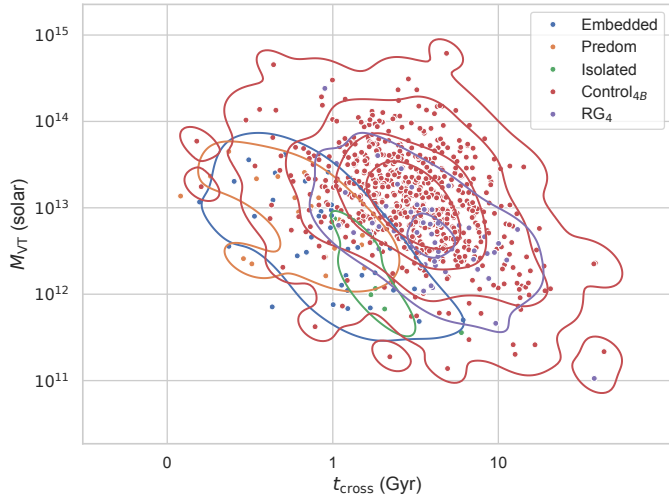


Fig. 16: Crossing time versus virial mass for the compact groups and their controls.

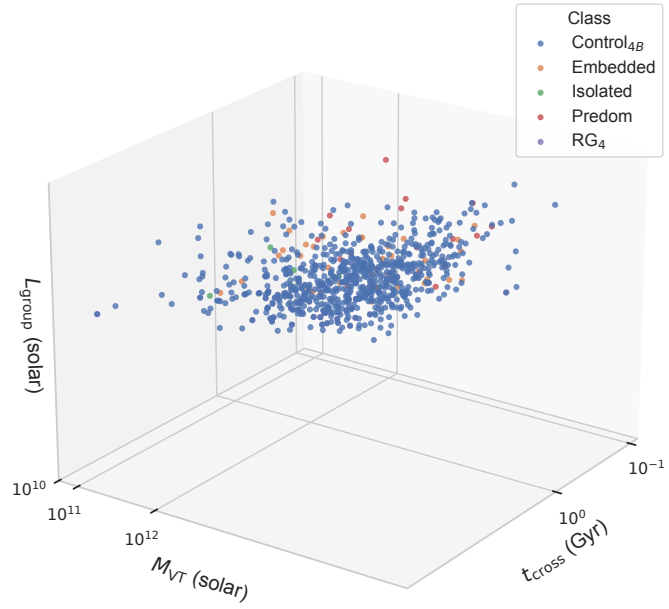


Fig. 17: Three-dimensional view of the crossing time, virial mass and group luminosity space.

of galaxies in different samples, as they do not necessarily have the same mass distribution. We will therefore compare colours properties in stellar mass bins.

Interpolation points for sSFR classification

Acknowledgements. We thank ...

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